

May 2024
CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS
(CS)

May 2024

Max. Marks: 100

Note:
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. In 8086 microprocessor , the address bus is _____ bit wide
A) 32 bit B) 8 bit
C) 16 bit D) 20 bit
2. The 16 bit flag of 8086 microprocessor is responsible to indicate _____
A) the condition of result of ALU operation B) the condition of memory
C) the result of addition D) the result of subtraction
3. The OF is called as _____
A) overdue flag B) overflow flag
C) one flag D) over flag
4. The BIU contains FIFO register of size 6 bytes called _____.
A) queue B) stack
C) segment D) register
5. If MN/MX is low the 8086 operates in mode
A) Minimum B) Maximum
C) Minimum and Maximum D) Medium
6. In 8086 microprocessor the following has the highest priority among all type interrupts?
A) DIV 0 B) TYPE 255
C) NMI D) OVERFLOW
7. Which of the following is not an arithmetic instruction?
A) INC (increment) B) CMP (compare)
C) DEC (decrement) D) ROL (rotate left)
8. The result of MOV AL, 65 is to store
A) store 0100 0010 in AL B) store 65H in AL
C) store 40H in AL D) store 0100 0001 in AL
9. ARM processors where basically designed for _____
A) Main frame systems B) Distributed systems
C) Mobile systems D) Super computers
10. When the CPU executes IRET
A) the control transfers from ISR to main program B) contents of IP and CS are retrieved
C) clears the trap flag D) clears the interrupt flag
11. What type of signal does the analogWrite() function output?
A) Pulse Width Modulated Signal B) Pulse Code Modulated Signal
C) Pulse Amplitude Modulated Signal D) Frequency Modulated Signal

12. What is the output of the program if the input to pin1 is 3.3V on the Arduino Due?

```
int pin1=A0;
void setup() {
  pinMode(pin1,INPUT);
  Serial.begin(9600);
}
void loop() {
  if(analogRead(pin1)==1023) {
    Serial.println("Digital HIGH Detected!");
  }
  else {
    Serial.println("Digital LOW Detected!");
  }
}
```

- A) Digital HIGH Detected
B) Digital LOW Detected
C) Runtime Error
D) Null
13. Example of an external interrupt is
A) divide by zero interrupt
B) keyboard interrupt
C) overflow interrupt
D) type2 interrupt
14. Which of the following instruction is not valid?
A) MOV AX, BX
B) PUSH AX
C) MOV AX, 5000H
D) MOV DS, 5000H
15. In general, the destination operand of an instruction can be
A) memory location
B) register
C) memory location and register
D) immediate data
16. In PUSH instruction, after each execution of the instruction, the stack pointer is
A) incremented by 1
B) decremented by 1
C) incremented by 2
D) decremented by 2
17. How many voltage levels are present when a pin uses digitalWrite()?
A) 1
B) 3
C) 2
D) 4
18. While CPU is executing a program, an interrupt exists then it
A) follows the next instruction in the program
B) jumps to instruction in other registers
C) breaks the normal sequence of execution of instructions
D) stops executing the program
19. How many arguments does the digitalWrite() function have?
A) 1
B) 2
C) 3
D) 4
20. The instruction that pushes the flag register on to the stack is
A) PUSH
B) PUSHF
C) POPF
D) POP

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Develop an ALP to search a key element in a list of 'n' 8-bit numbers using the Linear search algorithm. Display the position of the element if found, otherwise display the element not found.	8	L3	1	2
	b) How various Shift Instructions work? Differentiate between arithmetic and logical shift instructions. Explain with examples.	8	L2	1	1
2.	a) Develop an ALP with procedures to perform the following: i. Read a string from the keyboard terminated by carriage return ii. Display the string iii. Display the length of the string	8	L3	1	2
	b) Explain the flag register along with register format with respect to Intel 8086.	8	L2	1	1

- | | | | | | |
|-------|---|---|----|---|---|
| 3. a) | Develop an ALP to reverse a given string and check whether it is a palindrome or not. | 8 | L3 | 1 | 2 |
| b) | List and explain the various addressing modes available in Intel 8086. | 8 | L2 | 1 | 1 |

Unit – II

- | | | | | | |
|-------|--|---|----|---|-----|
| 4. a) | Explain the following 8086 Assembler Directives with an example for each: | | | | |
| i. | ASSUME | | | | |
| ii. | SEGMENT | | | | |
| iii. | INCLUDE | | | | |
| iv. | DB | 8 | L2 | 3 | 1 |
| b) | Describe the program status register of ARM processor with neat diagram. | 8 | L2 | 3 | 1,2 |
| 5. a) | Develop an ALP to read a string from the keyboard till carriage return and write into a file. | 8 | L3 | 2 | 1,2 |
| b) | Explain the ARM core model and its functional units with a neat diagram. | 8 | L2 | 4 | 1 |
| 6. a) | Develop an ALP procedure to convert two – digit Hexa – decimal to equivalent ASC11 values. | 4 | L3 | 2 | 1,2 |
| b) | Bring out the various differences between RISC and CISC machines. | 4 | L2 | 3 | 1,2 |
| c) | Develop an Embedded C program to read the input from 4x4 keypad and simulate operations as a calculator. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|---|
| 7. a) | Explain the following pin functions of 8086: | | | | |
| i. | NMI | 4 | L2 | 5 | 2 |
| ii. | READY | 6 | L1 | 5 | 2 |
| b) | Explain the Type 0 and Type 1 interrupts in 8086. | | | | |
| c) | Give the template for the MOV instruction of 8086 microprocessor. Generate machine code for the following instructions of Intel 8086 microprocessor. | | | | |
| i. | MOV CL, [BX] | 6 | L2 | 5 | 2 |
| ii. | MOV 43H[SI], DH | | | | |
| 8. a) | Differentiate between Maximum mode and minimum mode operation of 8086 microprocessor. | 6 | L2 | 5 | 2 |
| b) | What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. | 4 | L1 | 5 | 2 |
| c) | Explain the following maximum mode pin functions of 8086: | | | | |
| i. | QS0, QS1 | | | | |
| ii. | LOCK | | | | |
| iii. | Request/Grant | 6 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

May 2025

(For CS, CB)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Which index register in the 8086 is primarily used for operations involving string processing?
A) BP, SP
B) DH, DL
C) BH, BL
D) SI, DI
2. 8086 can access up to?
A) 512KB
B) 2Mb
C) 1Mb
D) 256KB
3. The instruction, MOV AX, 0005H belongs to the address mode
A) register
B) direct
C) immediate
D) register indirect
4. Match the following
a) DB 1) used to direct the assembler to reserve only 10-bytes
b) DT 2) used to direct the assembler to reserve only 4 words
c) DW 3) used to direct the assembler to reserve byte or bytes
d) DQ 4) used to direct the assembler to reserve words
A) a-3, b-2, c-4, d-1
B) a-2, b-3, c-1, d-4
C) a-3, b-1, c-2, d-4
D) a-3, b-1, c-4, d-2
5. Which group of instructions do not affect the flags?
A) Arithmetic operations
B) Logic operations
C) Data transfer operations
D) Branch operations
6. The instruction PUSHF is used to:
A) Push a register onto the stack
B) Push the flag register onto the stack
C) Pop a register from the stack
D) Load a segment register
7. If AX = 1001 0110 1010 1100B and CF = 1, what will be the result of RCL AX, 1?
A) 0010 1101 0101 1001B, CF = 1
B) 0010 1101 0101 1000B, CF = 1
C) 0010 1101 0101 1001B, CF = 0
D) 1001 0110 1010 1101B, CF = 1
8. The instruction RET is used to:
A) Return from subroutine
B) Jump to a memory address
C) Load a segment register
D) Move to the next instruction
9. Which function of INT 16H checks if a key is available?
A) AH = 00H
B) AH = 01H
C) AH = 02H
D) AH = 10H
10. What is the ASCII value of the hexadecimal digit 0x9?
A) 0x39
B) 0x49
C) 0x09
D) 0x1001

11. Set time interrupt from DOS interrupt will store the hour information in which register
 A) CH B) CL
 C) DH D) DL
12. In a program, a Macro is being called 'n' times. Then how many times is the machine code generated for the same?
 A) 1 time B) 'n' times
 C) 'n-1' times D) 'n+1' times
13. Which register in ARM holds the flags indicating the status of the processor?
 A) General-purpose register B) Program Counter (PC)
 C) Stack Pointer (SP) D) Program Status Register (PSR)
14. How does the 7-segment display represent numbers?
 A) Using rows and columns of LEDs B) By turning on specific segments of LEDs
 C) Through alphanumeric LCD cells D) By activating relays
15. Which built-in Arduino function is used for serial communication?
 A) digitalWrite() B) Serial.begin()
 C) analogRead() D) pinMode()
16. What is the primary design philosophy of ARM embedded systems?
 A) Complex Instruction Set Computing (CISC) B) Reduced Instruction Set Computing (RISC)
 C) Very Long Instruction Word (VLIW) D) Superscalar Processing
17. Which pin is used by the 8086 to indicate that an external device is ready to exchange data?
 A) READY B) INTR
 C) ALE D) HLDA
18. Which bit in the MOV instruction encoding specifies whether the data transfer is of a byte or a word?
 A) D (direction) bit B) MOD field bit
 C) W (word) bit D) REG field bit
19. During an interrupt response, which register is the first to be pushed onto the stack by the 8086?
 A) CS (Code Segment) B) IP (Instruction Pointer)
 C) FLAGS D) DS (Data Segment)
20. During a read cycle, when are the address lines present on the multiplexed bus?
 A) After the data is transferred B) During the WR signal activation
 C) When the ALE signal is active D) When the HLDA signal is active

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Identify the addressing modes of the instructions given below and justify the answer with clear explanation. i) MOV WORD PTR[SI], 50H ii) MOV DS:[1000H], 10H iii) MOV AX, Num[BX+DI]	6	L*3	1	2
	b) Write an Assembly level Program to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm.	6	L3	2	2
	c) Explain the use of instruction queue in 8086 architecture.	4	L2	1	1
2.	a) Describe the roles of various status flags and Control flags in Intel 8086 Microprocessor along with the flag register format. If the two numbers 1100H and 60C9H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.	6	L3	1	1

- b) How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example. 6 L1 2 1
- c) Write an Assembly level Program to display your name in front of the message "what is your name". 4 L3 2 2
3. a) Explain the working of following 8086 instructions along with example for each:
i) AAM ii) DAA 6 L2 2 1
- b) Explain the following assembler directives with example:
i. ASSUME
ii. EQU
iii. DD 6 L2 1 1
- c) Assume that SI is 3000H and the content of 2FFFH = 04H; content of 3000H = 4EH; Content of 3001H = 9FH. Which value is stored in AX register after execution of the statement MOV AX, [SI]. Justify your answer. 4 L3 2 2
- Unit – II**
4. a) Briefly explain the Embedded system software components. 8 L2 4 1
- b) Write an Embedded C program to control the speed of rotation of a stepper motor using potentiometer. 4 L3 4 2
- c) Explain the operation of the following DOS INT 21H functions:
i) Function 3AH
ii) Function 2CH 4 L2 3 1
5. a) With code snippet, explain Serial.begin(), Serial.available(), Serial.println() functions. 6 L2 4 1
- b) Differentiate between Macro and a Procedure. Write a procedure to read two digit BCD number from keyboard and store in BL register. 6 L3 4 1
- c) Compare and contrast RISC and CISC Architecture. 4 L2 3 1
6. a) With a basic layout diagram, explain the ARM program status register by highlighting the functionalities of different fields. 8 L2 4 1
- b) Write an Embedded C program to perform up-down counter on a 7-segment display using switches. 4 L3 4 2
- c) Write an Assembly Language Program to compute the factorial of a positive integer 'n' using a recursive procedure. 4 L3 3 2
- Unit – III**
7. a) Categorize the following signals into minimum mode signals and maximum mode signals. Write their roles.
(i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0' (iii) ALE (iv) DEN (v) HLDA 8 L1 5 1
- b) Explain the coding template for the MOV instruction of 8086 Intel microprocessor and Generate machine code for the following instructions of Intel 8086 microprocessor
MOV AL, [1000] [BX] 8 L3 5 2
8. a) List the 8086 signals having common functions in both minimum and maximum mode and explain any four signals in detail. 10 L1 5 1
- b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. 6 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

Scheme & Solution

Course Title :	MICROPROCESSOR AND EMBEDDED SYSTEMS	Course Code:	CS3005-1
Prepared by:	Dr. Anisha P Rodrigues	Set No.:	1
Faculty Signature:	<i>Aroba 18/10/25</i>	BOE Chairman Signature:	<i>J p k Selt</i>

Q.No.	Solution	Marks
	Unit – I	
1a)	Identify the addressing modes of the instructions given below and justify the answer with clear explanation. i) MOV WORD PTR[SI], 50H In this instruction, Destination uses register indirect addressing mode to specify destination location. Because the contents of SI register are used to generate effective address. Source uses Immediate Addressing mode ii) MOV DS:[1000H], 10H In this instruction, Destination uses direct addressing mode. Because the effective address is specified in the instruction. Source uses Immediate Addressing mode iii) MOV AX, Num[BX+DI] In this instruction, source uses base relative plus index addressing mode. Because base is provided by array name Num, BX register provides first index to array and the contents of DI acts as an offset Destination uses Register addressing mode	3X2
b)	Write an ALP to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm. <pre> DATA SEGMENT arr db 06H, 12H, 15H, 56H, 65H, 70H, 78H len db \$-arr mid db ? key db 79H msg1 db 10, 13, 'KEY NOT FOUND', 10, 13, '\$' msg2 db 10, 13, 'KEY FOUND AT POSITION \$' DATA ENDS CODE SEGMENT ASSUME CS: CODE, DS: DATA START: MOV AX, DATA OV DS, AX DEC len MOV DL, 00H MOV DH, len MOV BX, 0000 MOV CL, key </pre>	6


```

UP:
CMP DL, DH
JG NOTFOUND
MOV BL, DL
ADD BL, DH
SHR BL, 01H
MOV mid, BL
CMP CL, arr[BX]
JZ FOUND
JB FIRSTHALF
INC mid
MOV DL, mid
JMP UP
FIRSTHALF:
DEC mid
MOV DH, mid
JMP UP
NOTFOUND:
LEA DX, msg1
MOV AH, 09H
INT 21H
JMP EXIT
FOUND:
LEA DX, msg2
MOV AH, 09H
INT 21H
MOV BL, mid
CALL DISPHEXA
EXIT:
MOV AH, 4CH
INT 21H

DISPHEXA PROC NEAR
MOV DL, BL
MOV CL, 04H
SHR DL, CL
CMP DL, 09H
JBE L1
ADD DL, 07H
L1:
ADD DL, 30H
MOV AH, 02H
INT 21H
MOV DL, BL
AND DL, 0FH
CMP DL, 09H
JBE L2
ADD DL, 07H
L2:
ADD DL, 30H
MOV AH, 02H
INT 21H
RET
DISPHEXA ENDP
CODE ENDS
END START

```

c)

Explain the use of instruction queue in 8086 architecture.
 Explanation of the use of instruction queue in 8086 architecture.

- 2a) Describe the roles various status flags and Control flags in Intel 8086 Microprocessor along with the flag register format.
If the two numbers 1100H and 60C9H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.

Flag register format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
U	U	U	U	OF	DF	IF	TF	SF	ZF	U	AF	U	PF	U	CF

Explanation of Conditional flags, and Control flags.

Conditional flags: Direction Flag (DF), Interrupt Flag (IF), and Trap Flag (TF):

Explanation of these three

Conditional Flags: Overflow Flag (OF), Sign Flag (SF), Zero Flag (ZF), Auxiliary Carry Flag (AF)

Parity Flag (PF), Carry Flag (CF)

Explanation of these Six

If the two numbers 1100H and 60C9H are added gives 71C9H

- There is no carry from MSB. So Carry Flag is reset i.e., CF=0.
- There is no carry from bit 3 to bit 4. So Auxiliary Carry Flag is reset, i.e., AF=0.
- The result of addition is not zero. So Zero Flag is reset, i.e., ZF=0.
- The number of 1-bits is 8 which is even. So, Parity Flag is set, i.e., PF=1.
- The output is positive number with MSB 0 i.e., SF=0
- Overflow Flag is reset, i.e., OF=0.

So, CF=0, AF=0, ZF=0, PF=1, SF=0, OF=0

- b) How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example.

Difference between logical and arithmetic shift operations:

- The logical shifts move a 0 into the rightmost bit position for a logical left shift and a 0 into the leftmost bit position for a logical right shift. The arithmetic and logical left shifts are identical
- The arithmetic and logical right shifts are different because the arithmetic right shift copies the sign-bit through the number, while the logical right shift copies a 0 through the number
- Logical shift operations function with unsigned numbers, and arithmetic shifts function with signed numbers
- Logical shifts multiply or divide unsigned data, and arithmetic shifts multiply or divide signed data. A shift left always multiplies by 2 for each bit position shifted, and a shift right always divides by 2 for each bit position shifted

The arithmetic shift instructions are SAL, SAR explanation with example

The logical shift instructions are SHL, SHR explanation with example

- c) Write an ALP to display your name in front of the message "what is your name".

```
DATA SEGMENT
n DB ?
msg1 DB 10, 13, 'ENTER THE CHARACTER : $'
DATA ENDS
CODE SEGMENT
ASSUME CS: CODE, DS: DATA
START:
MOV AX, DATA
MOV DS, AX
```

Program

→ 4


```

LEA DX, msg1
MOV AH, 09H
INT 21H
MOV AH, 01H
INT 21H
MOV n, AL
MOV BL, n
CALL DISPHEXA
MOV AH, 01H
INT 21H
MOV AH, 4CH
INT 21H
DISPHEXA PROC NEAR
MOV DL, BL
MOV CL, 04H
SHR DL, CL
CMP DL, 09H
JBE L1
ADD DL, 07H
L1:
ADD DL, 30H
MOV AH, 02H
INT 21H
MOV DL, BL
AND DL, 0FH
CMP DL, 09H
JBE L2
ADD DL, 07H
L2:
ADD DL, 30H
MOV AH, 02H
INT 21H
RET
DISPHEXA ENDP
CODE ENDS
END START

```

3a) Explain the working of following 8086 instructions along with example for each:

i) AAM ii) DAA

Explanation along with example

(i) AAM:

This Instruction is used to convert from Hex Byte number to Unpacked BCD Number. This Instruction Divides the number in AL register by 0AH and stores the quotient in AH Register and remainder in AL register which is in BCD Format.

Example: If AL = 18H, after execution of AAM instruction, AH contains 02H and AL contains 04H.

(ii) DAA:

It is used after addition, to convert from hex number to packed BCD Number. This instruction performs as follows:

i) If the lower nibble of the sum in AL register is greater than 9 (i.e., from A to F) then DAA instruction adds 06 to the lower nibble. Else If the AF flag is high, then DAA instruction adds 06 to lower nibble.

ii) If the upper nibble of the sum in AL register is greater than 9 (i.e., from A to F) then DAA instruction adds 06 to the upper nibble. Else If the CF flag is high, then DAA instruction adds 06 to upper nibble.

b) Explain the following assembler directives with example:

- i. ASSUME
- ii. EQU

iii. DD

i) DD

- This directive DD defines a double word-type variable. The defined variable may have one or more values in the statement. If there is only one value, four bytes of consecutive memory locations are reserved. If there are multiple values, four bytes of memory locations are reserved for each value.
- The general format is:
Name of Variable DD Initial value or values
- Example:
NUMBER DD 23958634
The above directive informs assembler to reserve four bytes of memory locations for the variable named NUMBER and initialize with the number 23958634

3x2

ii) EQU

- The EQU or equate directive is used to assign names to constants used in programs
- Example: NUM EQU 07H
- Advantage of using EQU directive at start of programs is that if we want to modify the value of NUM it is enough if we change only at one place of declaration; it will reflect at all occurrences of NUM in the program

iii) ASSUME

- At any given time the 8086 works directly with only four physical segments: a code segment, a data segment, a stack segment and an extra segment
- The ASSUME directive tells the assembler which logical segment to use for each of these physical segments at a given time
- For example, the statement
- ASSUME CS:CODE, DS:DATA tells the assembler that the logical segment named CODE contains the instruction statements for the program and should be treated as a code segment
- It also tells the assembler that it should treat the logical segment DATA as data segment

c)

Assume that SI is 3000H and the content of 2FFFH = 04H; content of 3000H = 4EH; Content of 3001H = 9FH. Which value is stored in AX register after execution of the statement MOV AX [SI] Justify your answer.

The Value stored in CX register after execution of the statement MOV AX, [SI] is:
AH has 9FH and AL has 4EH.

Justification:

The value from specified location is always moved to Lower register (Here it is to AL) and the value from next higher addressed location is moved to higher register (Here it is to AH).

2+

2

Unit – II

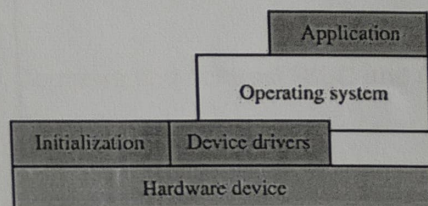
4a)

Briefly explain the Embedded system software components.

Diagram + explanation

An embedded system needs software to drive it

The following Figure shows four typical software components required to control an embedded device



Explanation of following components

- initialization code
- operating system
- device drivers

2+6

	Application The initialization code is the first code executed on the board and is specific to a particular target or group of targets It sets up the minimum parts of the board before handing control over to the operating system The operating system provides an infrastructure to control applications and manage hardware system resources The device drivers provide a consistent software interface to the peripherals on the hardware device An application performs one of the tasks required for a device ; For example, a mobile phone might have a diary application	
b)	Write an Embedded C program to control the speed of rotation of a stepper motor using potentiometer. <pre>const int stepPin = 3; const int dirPin = 2; int customDelay,customDelayMapped; // Defines variables void setup() { pinMode(stepPin,OUTPUT); pinMode(dirPin,OUTPUT); digitalWrite(dirPin,HIGH); //Enables the motor to move in a clockwise direction } void loop() { customDelayMapped = speedUp(); digitalWrite(stepPin, HIGH); delayMicroseconds(customDelayMapped); digitalWrite(stepPin, LOW); delayMicroseconds(customDelayMapped); } // Function for reading the Potentiometer int speedUp() { int customDelay = analogRead(A0); // Reads the potentiometer int newCustom = map(customDelay, 0, 1023, 1000,4000); return newCustom; }</pre>	4
c)	Explain the operation of the following DOS INT 21H functions: i) Function 3AH ii) Function 2CH i) Function 3AH Removes sub-directory using specified disk and path Calling parameters AH=3Ah DS:DX -> seg: offset of ASCIIZ path specification(ASCIIZ pathname of directory to be removed) Return: If successful, CF clear If not successful, CF set ,AX = error code ii) Function 2CH Get DOS system time Returns: CH = Hour (0 to 23) CL = Minutes (0 to 59) DH = seconds (0 to 59) DL = 100th of seconds (0 to 99)	2x2
5a)	With code snippet, explain Serial.begin(),Serial.available(),Serial.println() functions. Serial.begin(baud_rate): baud_rate : The baud rate that will be used for serial communication. Can be 4800, 9600, 14400, 19200, etc. This function is used to define the baud rate that will be used for serial communication. Example : Serial.begin(9600) defines 9600 baud rate for communication.	3x2

Serial.available():

This function is used to get the number of bytes available for reading from the serial port. It gives the number of bytes of data that has arrived and is stored in the serial receive buffer.

Example:

```
if(Serial.available())
```

If data available at serial port, take action.

Serial.println(value):

value : character, string, number to be printed.

This function is used to print data to a serial port in a form that is human readable (character, strings, numbers) followed by a carriage return (\r) and a new line character (\n).

Example:

```
Serial.println("hello")
```

b)

Differentiate between Macro and a Procedure. Write a procedure to read two digit BCD number from keyboard and store in BL register.

Difference between PROCEDURE & MACRO.

Procedure	Macro
Procedures are used for large group of instructions to be repeated.	Procedures are used for small group of instructions to be repeated.
Object code is generated only once in memory.	Object code is generated everytime the macro is called.
CALL & RET instructions are used to call procedure and return from procedure.	Macro can be called just by writing its name.
Length of the object file is less	Object file becomes lengthy.
Directives PROC & ENDP are used for defining procedure.	Directives MACRO and ENDM are used for defining MACRO
More time is required for it's execution	Less time is required for it's execution
Procedure can be defined as Procedure_name PROC ----- Procedure_name ENDP	Macro can be defined as MACRO-name MACRO [ARGUMENT ,____ ARGUMENT N] ----- ENDM
For Example Addition PROC near ----- Addition ENDP	For Example Display MACRO msg ----- ENDM

READBCD HEXA

MOV AH, 01H;first digit

INT 21H

MOV CL, 04

SHL AL, CL

MOV BL, AL

MOV AH, 01H;second digit

INT 21H

AND AL, 0FH

ADD AL, BL

RET

READBCD ENDP

c)

Compare and contrast RISC and CISC Architecture.

CISC	RISC
Complex instructions, taking multiple clock	Simple instructions, taking single clock
Emphasis on hardware, complexity is in the micro-program/processor	Emphasis on software, complexity is in the Compiler
Complex instructions, instructions executed by micro-program/processor	Reduced instructions, instructions executed by Hardware
Variable format instructions, single register set and many instructions	Fixed format instructions, multiple register sets and few instructions
Many instructions and many addressing modes	Fixed instructions and few addressing modes
Conditional jump is usually based on status register bit	Conditional jump can be based on a bit anywhere in memory
Memory reference is embedded in many instructions	Memory reference is embedded in LOAD/STORE instructions

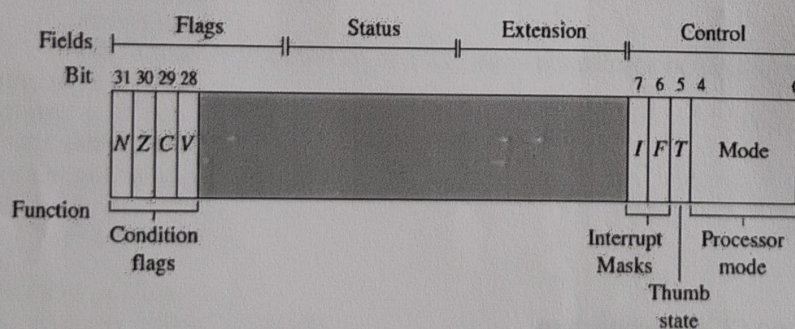
6a) With a basic layout diagram, explain the ARM program status register by highlighting the functionalities of different fields.

Basic layout diagram + Explanation of ARM program status register by highlighting the functionalities of different fields

2+6

Basic layout of a generic program status register

- The ARM core uses the cpsr to monitor and control internal operations
- The cpsr is a dedicated 32-bit register and resides in the register file
- The following Figure shows the basic layout of a generic program status register
- Note that the shaded parts are reserved for future expansion



The cpsr is divided into four fields, each 8 bits wide: flags, status, extension, and control

Explanation of following fields

- Flags
- Status
- Extension
- control

b) Write an Embedded C program to perform up-down counter on a 7-segment display using switches.

```
#include <TM1637Display.h>
#define switch_1 14
#define switch_2 15
```



```

const int CLK = 8; //Set the CLK pin connection to the display
const int DIO = 9; //Set the DIO pin connection to the display
const uint8_t blank[] = {0x00, 0x00, 0x00, 0x00};
TM1637Display display(CLK, DIO); //set up the 4-Digit Display.
int num = 0;
bool UP = 0;
bool DOWN = 0;

```

```

void setup(){
  pinMode(switch_1, INPUT);
  pinMode(switch_2, INPUT);
  display.setBrightness(7); //set the display to maximum brightness
  display.setSegments(blank); //clear display
}

```

```

void loop(){
  call();
  if(UP){
    if(num == 9999)
      num = 0;
    call();
    num++; // increment 'num'
  }
  else if(DOWN){
    if(num == 0)
      num = 9999;
    call();
    num--; // decrement 'num'
  }
  else{
    num = 0;
  }
}

```

```

void call(){
  display.showNumberDec(num, true, 4, 0);
  delay(500); //wait 200 milliseconds
  if(digitalRead(switch_1)){ // When switch_1 is pressed
    UP = 1;
    DOWN = 0;
  }
  if(digitalRead(switch_2)){ // When switch_2 is pressed
    UP = 0;
    DOWN = 1;
  }
}

```

c) Write an Assembly Language Program to compute the factorial of a positive integer 'n' using a recursive procedure.

```

DATA SEGMENT
n DB 06H
fact DW ?
msg DB 'FACTORIAL(6)=$'
DATA ENDS

```

```

CODE SEGMENT
ASSUME CS:CODE, DS:DATA
START:
MOV AX, DATA
MOV DS, AX

```



```

MOV AX,1
MOV BL,n
MOV BH,00
CALL FACTORIAL
MOV fact,AX
MOV BL, AH
LEA DX,msg
MOV AH,09H
INT 21H
CALL DISPHEXA
MOV BX,fact
CALL DISPHEXA
MOV AH,4CH
INT 21H
FACTORIAL PROC
CMP BX,1
JE L1
PUSH BX
DEC BX
CALL FACTORIAL
POP BX
MUL BX
L1:
RET
FACTORIAL ENDP

```

```

DISPHEXA PROC
MOV DL,BL
MOV CL,04
SHR DL,CL
CMP DL,09H
JBE L2
ADD DL,07H
L2:
ADD DL,30H
MOV AH,02H
INT 21H
MOV DL,BL
AND DL,0FH
CMP DL,09H
JBE L3
ADD DL,07H
L3:
ADD DL,30H
MOV AH,02H
INT 21H
RET
DISPHEXA ENDP
CODE ENDS
END START

```

Unit – III

7a) Categorize the following signals into minimum mode signals and maximum mode signals. Write their roles.

(i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0' (iii) ALE (iv) DEN (v) HLDA

Categorization + Explanation

Categorization:

Minimum Mode Signals: (iii) ALE (iv) DEN (v) HLDA

Maximum Mode Signals: (i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0

1+7

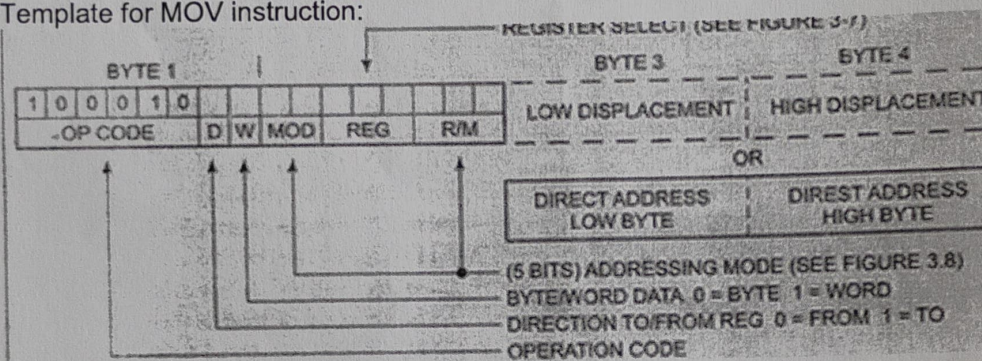
Explanation:

- (i) ALE: Address Latch Enable shows that the 8086 address/data bus contains address information.
- (ii) DEN: Data Bus Enable activates external data bus buffers.
- (iii) HLDA: Hold Acknowledge indicates that 8086 has entered the hold state.
- (iv) S2/ to S0/ The Status bits indicate the function of the current bus cycle. These signals are normally decoded by the 8288 bus controller.
- (v) RQ/ / GT1/ and RQ/ / GT0/ The Request/Grant Signals requests the direct memory access during maximum mode operation. These are bidirectional and are used to both request and grant a DMA operation.

b) Explain the coding template for the MOV instruction of 8086 Intel microprocessor and Generate machine code for the following instructions of Intel 8086 microprocessor
MOV AL, [1000] [BX]

Template explanation + Machine code generation

Template for MOV instruction:



Explanation of OPCODE, D, W, MOD, REG, R/M, displacement

MOV AL, [1000] [BX]

Machine code: 1000 1010 10000 111 00000000 00100000=8A870010 H

8a) List the 8086 signals having common functions in both minimum and maximum mode and explain any four signals in detail.

Listing the signals used in both minimum and maximum mode operation + explanation of any 4 signals

List of signals used in both minimum and maximum mode operation

- AD0 - AD15 (I/O)
- A19/S6, A18/S5, A17/S4, A16/S3 (O)
- BHE/S7 (O)
- RD (O)
- TEST (I)
- INTR (I)
- NMI (I)
- Reset (I)
- Ready (I)
- CLK (I)
- MN/MX (I)

b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling.

Explanation of interrupt vector and vector table + steps

Various steps involved in interrupt handling:

- It decrements SP by 2 and pushes the flag register on the stack
- Disables INTR by clearing the IF

- It resets the TF in the flag Register
- It decrements SP by 2 and pushes CS on the stack
- It decrements SP by 2 and pushes IP on the stack
- Fetch the ISR address from the interrupt vector table
